TYPE E EXAMPLES BEYOND THE λ-MINUSCULE RANGE: THE EXTREMAL WEIGHTS WITH A c-TRACE ENTRY 3

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ABSTRACT. Let Γ be a simply-laced Dynkin diagram, $\lambda = \omega_\kappa$ a fundamental weight, w an element of the Weyl group of minimal length in wW_λ, and V the unique T-fixed representation on $\mathfrak{M}(\lambda - w\lambda, \lambda)$. The *$c$-trace* of w is the sequence $c_p = \langle \alpha_{i_p}^\vee, \mu^{(p-1)} \rangle$ attached to a reduced word, and w is λ-minuscule exactly when every $c_p = 1$. We determine, by an exhaustive computation over all of E_6 and E_7 and all fundamental λ, every extremal weight whose c-trace attains the value 3 and whose V has simple spectrum. There are exactly nine, listed in Table 1; one of them, number (4), is not fully commutative. No extremal weight in $E_6 \cup E_7$ with a c-trace entry ≥ 4 has simple spectrum. The computation is carried out twice, by independent methods, and the claims of this note are computational, not proved.

1. WHAT IS COMPUTED, AND WHAT IS NOT

The table and the counts in this note are the output of a finite computation, performed in two independent ways that agree on every case where both terminate. We say this at the outset because the companion work is proof-based and we do not wish the two to be confused. The one exception is §4, which is proved, and which is what explains the position of the weight-3 step; each statement there is labelled as proved or as computed.

The two methods are:

(i) *Socle filtration.* Build the injective Π-module I_κ explicitly as $\Pi e_{\nu(\kappa)}$, where ν is the Nakayama permutation, and compute the iterated socle $\mathrm{soc}_\sigma(I_\kappa)$ along the c-weighted sequence σ of a reduced word for w. Each socle step of colour j is the kernel computation

$$X \longmapsto \{\, x \in I_\kappa : \ a_{j\to i}\, x \in X \text{ for every } i \sim j \,\},$$

so the whole filtration costs one linear-algebra pass per letter. By the main theorem of the companion work, $V \cong \mathrm{soc}_\sigma(I_\kappa)$, so the graded dimensions of the socle layers are the graded dimensions of V, and simple spectrum is the condition that every layer be multiplicity-free in each colour.

(ii) *Submodule search.* Search directly for a graded submodule of I_κ of dimension vector $v = \lambda - w\lambda$, by depth-first search over the subspace lattice degree by degree. This uses no theorem; it is slow, and in E_7 it does not terminate on the larger weights.

We validated (i) against (ii) on 1001 extremal weights — all of $A_3\,\omega_1$, $D_4\,\omega_2$, $D_5\,\omega_4$, $E_6\,\omega_4$, $E_6\,\omega_5$ and $E_6\,\omega_6$ — with no disagreement: the dimension vectors agree in every case, and the socle layers agree with the layers of (ii) after reversing the degree, which is the expected shift between a projective generated in degree 0 and an injective with socle in degree 0. The validation set was chosen to include vertices *not* fixed by ν (A_3, $D_5\,\omega_4$, $E_6\,\omega_5$, $E_6\,\omega_6$), since at a ν-fixed vertex the distinction between Πe_κ and $\Pi e_{\nu(\kappa)}$ is invisible and an error in the Nakayama bookkeeping would not show.

Method (i) alone was then run exhaustively. We note precisely what "exhaustively" covers:

- E_6: all six fundamental λ, all 1272 nontrivial extremal weights (orbit sizes $27, 72, 216, 720, 216, 27$). Complete.
- E_7: all seven fundamental λ, all 17635 nontrivial extremal weights; the orbit sizes are

$$126,\ 576,\ 2016,\ 10080,\ 4032,\ 756,\ 56.$$

  Complete.
- E_8: $\lambda = \omega_1$ and $\lambda = \omega_8$ only (orbit sizes 2160 and 240). Both reach $\max_p c_p = 2$ and contain no element with a c-trace entry 3 at all. The remaining six fundamental weights of E_8 have orbits too large for the present implementation and are *not* covered.

2. THE NINE EXAMPLES

Convention 2.1. Vertices are numbered as in Bourbaki: E_n has the chain $1-3-4-5-\cdots-n$ with vertex 2 attached to vertex 4, so vertex 4 is trivalent. A weight μ is written in the basis of fundamental weights,

$\mu = (\langle \alpha_1^\vee, \mu\rangle, \dots, \langle \alpha_n^\vee, \mu\rangle)$. Dimension vectors v are written in the basis of simple roots, so $\lambda - w\lambda = \sum_i v_i \alpha_i$. A reduced word $s_{i_1} \cdots s_{i_\ell}$ acts with s_{i_ℓ} first, and the c-trace is listed in that order, so c_1 is the entry of the rightmost letter. The displayed V lists the socle layers of $\mathrm{soc}_\sigma(I_\kappa)$ from the bottom up, each layer as the multiset of its colours.

Theorem 2.2 (computational)**.** *Let $\Gamma \in \{E_6, E_7\}$, let $\lambda = \omega_\kappa$ be any fundamental weight and let w be of minimal length in wW_λ. Suppose the c-trace of w attains the value 3. Then T acts on V with simple spectrum if and only if (Γ, λ, w) is one of the nine triples of Table 1. Of the 7094 such triples in $E_6 \cup E_7$, these nine are the only ones.*

	Γ, λ	w	$\ell(w)$	c-trace
(1)	E_6, ω_4	$s_2s_4s_5s_6s_3s_4s_5s_1s_3s_4$	10	$(1^9, 3)$
(2)	E_7, ω_3	$s_3s_4s_5s_6s_7s_2s_4s_5s_6s_1s_3s_4s_5s_2s_4s_3$	16	$(1^6, 2, 1^8, 3)$
(3)	E_7, ω_3	$s_1s_3s_4s_5s_6s_7s_2s_4s_5s_6s_3s_4s_5s_2s_4s_3$	16	$(1^{15}, 3)$
(4)	E_7, ω_3	$s_1s_3s_4s_5s_6s_7s_2s_4s_5s_6s_1s_3s_4s_5s_2s_4s_3$	17	$(1^6, 2, 1^8, 3, 1)$
(5)	E_7, ω_4	$s_2s_4s_5s_6s_3s_4s_5s_1s_3s_4$	10	$(1^9, 3)$
(6)	E_7, ω_4	$s_7s_2s_4s_5s_6s_3s_4s_5s_1s_3s_4$	11	$(1^9, 3, 1)$
(7)	E_7, ω_4	$s_6s_7s_2s_4s_5s_6s_3s_4s_5s_1s_3s_4$	12	$(1^9, 3, 1^2)$
(8)	E_7, ω_4	$s_5s_6s_7s_2s_4s_5s_6s_3s_4s_5s_1s_3s_4$	13	$(1^9, 3, 1^3)$
(9)	E_7, ω_5	$s_2s_4s_5s_6s_7s_3s_4s_5s_6s_1s_3s_4s_5$	13	$(1^{12}, 3)$
	$\mu = w\lambda$	$v = \lambda - w\lambda$	$\lvert v\rvert$	V, bottom layer first
(1)	$(0,-3,0,2,0,0)$	$(1,3,2,3,2,1)$	12	$\{2\}\,\{4\}\,\{2,3,5\}\,\{1,4,6\}\,\{2,3,5\}\,\{4\}$
(2)	$(1,0,-3,2,0,0,0)$	$(2,2,5,4,3,2,1)$	19	$\{3\}\,\{4\}\,\{2,3,5\}\,\{1,4,6\}\,\{3,5,7\}\,\{4,6\}\,\{2,3,5\}\,\{1,4\}\,\{3\}$
(3)	$(-3,0,2,0,0,0,0)$	$(3,2,3,4,3,2,1)$	18	$\{1\}\,\{3\}\,\{4\}\,\{2,5\}\,\{1,4,6\}\,\{3,5,7\}\,\{4,6\}\,\{2,5\}\,\{1,4\}\,\{3\}$
(4)	$(-1,0,-2,2,0,0,0)$	$(3,2,5,4,3,2,1)$	20	$\{1\}\,\{3\}\,\{4\}\,\{2,3,5\}\,\{1,4,6\}\,\{3,5,7\}\,\{4,6\}\,\{2,3,5\}\,\{1,4\}\,\{3\}$
(5)	$(0,-3,0,2,0,0,1)$	$(1,3,2,3,2,1,0)$	12	$\{2\}\,\{4\}\,\{2,3,5\}\,\{1,4,6\}\,\{2,3,5\}\,\{4\}$
(6)	$(0,-3,0,2,0,1,-1)$	$(1,3,2,3,2,1,1)$	13	$\{2\}\,\{4\}\,\{2,3,5,7\}\,\{1,4,6\}\,\{2,3,5\}\,\{4\}$
(7)	$(0,-3,0,2,1,-1,0)$	$(1,3,2,3,2,2,1)$	14	$\{2\}\,\{4,6\}\,\{2,3,5,7\}\,\{1,4,6\}\,\{2,3,5\}\,\{4\}$
(8)	$(0,-3,0,3,-1,0,0)$	$(1,3,2,3,3,2,1)$	15	$\{2,5\}\,\{4,6\}\,\{2,3,5,7\}\,\{1,4,6\}\,\{2,3,5\}\,\{4\}$
(9)	$(0,-3,0,2,0,0,0)$	$(1,3,2,3,3,2,1)$	15	$\{2\}\,\{4\}\,\{2,3,5\}\,\{1,4,6\}\,\{2,3,5,7\}\,\{4,6\}\,\{5\}$

TABLE 1. The nine extremal weights in $E_6 \cup E_7$ whose c-trace attains 3 and whose T-fixed representation has simple spectrum. All nine have exactly one entry $c_p = 3$. Numbers (2) and (4) have in addition a single entry 2; the other seven are minuscule away from the single 3. Number (4) is the only one that is not fully commutative.

Remark 2.3 (what the table does and does not pattern)**.** Three features are worth recording, and one tempting guess is false.

(a) The 3 sits at or near the end, but "at the end" is not the criterion. Four of the nine have c-trace exactly $(1, \dots, 1, 3)$. It is tempting to conjecture that among the elements with $\max_p c_p = 3$, simple spectrum holds precisely for those with c-trace $(1, \dots, 1, 3)$. This is false: of the nine, five have another shape, and conversely the condition is not sufficient in the $\max_p c_p = 2$ range, where 15 elements of $E_6 \cup E_7$ have c-trace $(1, \dots, 1, 2)$ and fail simple spectrum. We flag this because the guess survives any inspection of minimal-length representatives — in each of the three families above, the shortest element does have c-trace $(1, \dots, 1, 3)$ — and is refuted only by the full list.

(b) λ is always adjacent to or at the trivalent node. All nine occur at $\lambda \in \{\omega_3, \omega_4, \omega_5\}$, and in E_6 only at ω_4. This is consistent with the E_8 data: ω_1 and ω_8, the two fundamental weights farthest from vertex 4, admit no c-trace entry 3 whatsoever.

(c) Items (5)–(8) and (9) extend (1) by minuscule letters. Example (1) is the unique one in E_6; (5) is its image in E_7, with the same w, the same v in the first six coordinates and the same V; and (6), (7), (8) are obtained from (5) by prepending s_7, s_6s_7, $s_5s_6s_7$, each step appending a single minuscule letter and adding one box to V. So the nine are not nine unrelated accidents but four orbits of one phenomenon: a single non-minuscule step of weight 3, surrounded by minuscule steps.

(d) The weight-3 step is never at the trivalent node. Reading the c-trace in the order fixed by the Convention above, its last entry belongs to the leftmost letter of w, so the table already locates the 3: it sits at vertex 2 in six of the nine, at vertex 3 in two, and at vertex 1 in one. Vertex 4 never carries it. More than that, in all nine the trivalent node carries $c_p = 1$ at *every* one of its occurrences. That is not an accident of the list, and §4 says why.

(e) $c_p \geq 4$. There are 7560 extremal weights in $E_6 \cup E_7$ with $\max_p c_p \geq 4$ — all of them in E_7 at $\lambda = \omega_4$, all with $\max_p c_p = 4$ exactly — and not one has simple spectrum. We state this as a computation and not as a theorem; it is the natural next thing to prove, and a proof would have to explain why 3 is possible and 4 is not.

3. Example (1) in full

The smallest of the nine is small enough to be checked by hand, and we give it in full so that a reader can verify Table 1 independently of any code.

Take $\Gamma = E_6$, $\lambda = \omega_4$ and $w = s_2s_4s_5s_6s_3s_4s_5s_1s_3s_4$, of length 10. Acting with the rightmost letter first on $\lambda = (0,0,0,1,0,0)$, the trace is

p	i_p	c_p	$\mu^{(p)}$
1	4	1	$(0,1,1,-1,1,0)$
2	3	1	$(1,1,-1,0,1,0)$
3	1	1	$(-1,1,0,0,1,0)$
4	5	1	$(-1,1,0,1,-1,1)$
5	4	1	$(-1,2,1,-1,0,1)$
6	3	1	$(0,2,-1,0,0,1)$
7	6	1	$(0,2,-1,0,1,-1)$
8	5	1	$(0,2,-1,1,-1,0)$
9	4	1	$(0,3,0,-1,0,0)$
10	2	3	$(0,-3,0,2,0,0)$

so $\mu = w\lambda = (0,-3,0,2,0,0)$ and the single non-minuscule step is the last one, at the trivalent neighbour 2, where $\langle \alpha_2^\vee, \mu^{(9)} \rangle = 3$. Hence

$$v \;=\; \lambda - w\lambda \;=\; \alpha_1 + 3\alpha_2 + 2\alpha_3 + 3\alpha_4 + 2\alpha_5 + \alpha_6, \qquad |v| = 12.$$

Since $\mu \in W\lambda$ we have $(\mu,\mu) = (\lambda,\lambda)$, so $\delta(v) = 0$ and v is extremal; by the uniqueness statement of the companion work, I_κ has at most one submodule of dimension vector v, and exhibiting one identifies it with V. The weighted sequence σ is the word read in trace order with the last letter tripled,

$$\sigma \;=\; (4,3,1,5,4,3,6,5,4,2,2,2),$$

and the iterated socle $\mathrm{soc}_\sigma(I_4)$ has layers, from the bottom,

$$\{4\}, \quad \{2,3,5\}, \quad \{1,4,6\}, \quad \{2,3,5\}, \quad \{4\}, \quad \{2\}.$$

Summing the layers gives v, each layer is multiplicity-free, and each colour appears in distinct layers; so the spectrum is simple. Note that the last socle step, of colour 2 and multiplicity 3, contributes the three occurrences of colour 2 in three *different* layers — the second, fourth and sixth from the bottom — which is exactly why a step of weight 3 need not destroy simple spectrum. A step of weight c contributing c boxes to a *single* layer is the failure mode, and it is what happens at all of the other 479 elements of E_6 with a c-trace entry 3.

4. Why the weight-3 step avoids the node

Two statements of the companion work bear directly on the table, and unlike the rest of this note they are proved. Write t for the trivalent node.

Lemma 4.1 (proved)**.** *$v_i = \sum_{p:\, i_p = i} c_p$ for every vertex i.*

This is the telescoping of $\mu^{(p)} = \mu^{(p-1)} - c_p\alpha_{i_p}$ against $\lambda - w\lambda = \sum_i v_i\alpha_i$.

Lemma 4.2 (proved)**.** *Suppose every connected component of $\Gamma \setminus \{t\}$ is of type A — which holds for D_n and for E_6, E_7, E_8, hence for every simply-laced Γ with a branch node. If t occurs exactly once in a reduced word for w, its c-trace entry is* 1*.*

The proof is short. The suffix u after the unique occurrence of t lies in the parabolic $W_{I\setminus\{t\}}$, and writing $\lambda - u\lambda = \sum_j v_j(u)\alpha_j$ and pairing with $\alpha_t^\vee$ gives $c_p = \langle \alpha_t^\vee, \lambda \rangle + \sum_{j\sim t} v_j(u)$. The components of $\Gamma \setminus \{t\}$ not containing κ fix λ, so only one term survives; and in a type-A component the coefficient at the end of the chain adjacent to t is at most 1. As a check, enumerating the whole parabolic — of order 72, 288, 1440 for E_6, E_7, E_8 — confirms that the maximum over every possible lone node step is 1, for every fundamental λ.

Together these give, *in type D*, that a c-trace entry ≥ 2 at t forbids simple spectrum: by Lemma 4.2 the colour t then occurs at least twice, so $v_t \geq 3$ by Lemma 4.1, and the type-D criterion $v_t \leq 2$ denies simple spectrum. In type E the last step is unavailable, since $v_t \geq 3$ does *not* forbid simple spectrum there. What we can report is that the conclusion holds anyway: among the 1272 extremal weights of the six fundamental λ of E_6, the 831 whose c-trace exceeds 1 at the trivalent node all fail to have simple spectrum, with no exception. We do not know a proof of the type-E statement.

This is the reason Remark 2.3(d) looks the way it does: a weight-3 step cannot sit at t, so it is forced onto a vertex of smaller degree, and the table shows it at vertices 2, 3 and 1.

Lemma 4.1 also yields a bound where the orbits cannot be enumerated. Simple spectrum means every colour-i column of V is multiplicity free, so $v_i \leq L_i$, the number of degrees in which the colour-i column of I_κ is nonzero; and $c_p \leq v_{i_p}$ by Lemma 4.1, every c_q being positive. Hence

$$c_p \;\leq\; L_{i_p} \qquad \text{for every } p,$$

whenever V has simple spectrum. Computing the L_i from the profile of I_κ — which needs no orbit enumeration — gives $c_p \leq 6$ in E_6, $c_p \leq 9$ in E_7 and $c_p \leq 15$ in E_8. These are far from sharp: the observed maximum of c_p over *all* extremal weights is 3 in E_6 and 4 in E_7, and over the simple-spectrum ones it is 3 in both. But they are proved, they hold in E_8, and they reduce Remark 2.3(e) from an unbounded statement to a finite one.

5. Reproducing the computation

The scripts are in the repository accompanying this note. With `rootsys.py`, `proj2.py`, `wordtrace.py`, `socle.py` and `dfs3.py` in the working directory:

- `python3 socle.py` — validates the socle method against the submodule search on the worked examples, including the D_4 element that is fully commutative without simple spectrum.
- `python3 socval.py E6 5` — full cross-validation of the two methods over one orbit; likewise `A3 1`, `D5 4`, `E6 6`, `E6 4`, `D4 2`.
- `python3 escan.py E6` and `python3 escan.py E7` — the exhaustive scans, about 3 seconds and about 3 minutes respectively.
- `python3 enine.py` — regenerates Table 1.
- `python3 escan.py E8 1 1` and `python3 escan.py E8 8 8` — the two E_8 weights that are within reach.

All computations are linear algebra over $\mathbb{F}_p$, with p set by the environment variable `DKKP` (default 7). The socle filtration is a sequence of kernels of integer matrices, so a p-dependent answer would require a p-torsion coincidence in one of those kernels. We ran the complete $E_6 \cup E_7$ scan at $p = 7$, $p = 11$ and $p = 32003$; all four counts of §1 and Remark 2.3, and the list of nine itself, were identical at all three. This is evidence of p-independence, not a proof of it.

6. Open ends

(1) E_8 at $\lambda = \omega_2, \ldots, \omega_7$ is not covered. The obstruction is the orbit enumeration, not the socle computation: the latter costs about 10–100 ms per element, so ω_7 (orbit 6720), ω_2 (orbit 17280) and ω_6 (orbit 60480) are a matter of minutes to hours, while $\omega_3, \omega_4, \omega_5$ run to the millions and need the search restricted by the c-trace rather than run over the whole orbit.

(2) Remark 2.3(e) — that $c_p \geq 4$ never admits simple spectrum — is unproved, and is the sharpest statement the data suggests. §4 reduces it to a finite check by bounding c_p above, but the bound is 6, 9, 15 in E_6, E_7, E_8 against a true value of 3, so the gap is still wide.

(3) A conceptual description of the nine, replacing the list. Remark 2.3(c) reduces them to four, and the mechanism isolated at the end of §3 — whether the c boxes of a weight-c step land in c distinct socle layers — is presumably the right invariant to make precise.

(4) Whether the nine examples survive in characteristic 2, where the strand decomposition of the companion work is unavailable.

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